

B2W Ops API Help Guide

How to reach a customer’s Ops API, get the right credentials, generate a bearer token in Postman, make your first calls, and work API tickets. Every example uses the **B2WTechSupport** environment.

● Example environment

Checked Oct 2, 2026

ENVIRONMENT	B2WTechSupport
CLUSTER	b2w-eus10.b2w.trimble.com
OPS API URL	https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport
API VERSION	26.1.2.1
TRIMBLE ID MODE	On IsTIDEnabled: true

Understand the basics

What the Ops API is, what every call is made of, and how to find any customer's API and its documentation.

1 What is the B2W Ops API?

SHORT ANSWER

The Ops API is how other software reads and writes B2W Ops data, such as jobs, employees, and equipment, without anyone opening an Ops screen. It is a REST API that sends and receives JSON, and every customer environment has its own.

An API (application programming interface) is a published way for programs, not people, to ask another system to do something over a network. The training used three pieces of the same building to explain it:



OPS DATABASE

The filing cabinet

Stores the raw data and permanent records behind the scenes.



OPS WEBSITE

The front desk

Screens that people use to find and change records.



OPS API

The service window

A written menu that outside software uses to request records.

OUTSIDE SYSTEMS

ERP · telematics · B2W imports

Send requests to the API.



OPS API

Enforces Ops security rules

Checks permissions and validation on every call.



OPS DATABASE

System of record

Only changed through those rules.

Customers use it to bring data in from ERP systems, pull data out for reporting dashboards, connect field telematics, and automate repeatable tasks. B2W uses the same API internally for native job imports and true-up batches.



The API never goes around Ops security

It enforces the same security permissions and database validation rules as the application. Nothing bypasses them to write to the database directly, so Ops stays the system of record.

What can you do with it?

Anything you can do in the application, for the functional areas the API exposes, within the permissions of whoever signed in. The B2WTechSupport catalog lists 63 functional areas and 272 operations (see [question 16](#)).

Ops API vs. legacy Web Services

Customers sometimes still use the older SOAP Web Services. They are a different product with a different support path, so check which one you are looking at first.

	OPS API 2.0 (REST)	LEGACY WEB SERVICES (SOAP)
Address	One URL per resource, such as <code>/Employee</code>	<code>.asmx</code> service URLs
Methods	<code>GET</code> <code>POST</code> <code>PUT</code> <code>DELETE</code>	<code>POST</code> for every call
Data format	JSON	XML envelopes, with the operation name inside
Querying	OData: <code>\$filter</code> , <code>\$select</code> , <code>\$orderby</code> , <code>\$top</code> , <code>\$skip</code>	Parameters inside the XML
Page size	Up to 100 records per GET	Not applicable



Steer customers to the Ops API

Use the Ops API wherever you can. In the training example, adding 10 employee records took 10 separate legacy SOAP calls, while the Ops API handled the same work in one call.

2

What makes up an API call?

SHORT ANSWER

Every call has an **address** (the URL), a **method** (GET, POST, PUT, or DELETE), **headers** (including who you are), and for create and update calls, a JSON **body**. Every response comes back with a status code and, usually, JSON data.

PART	WHAT IT DOES	EXAMPLE
1. Address	Where the request goes: the customer's server and the resource	<code>https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport/Employee</code>

PART	WHAT IT DOES	EXAMPLE
2. Method	The action to take	GET read · POST create · PUT update · DELETE remove
3. Headers	Extra instructions, including who is calling	Authorization: Bearer <token> Content-Type: application/json
4. Body	The data you send, as JSON, when you create or update	{ "LastName": "Newman", ... }

The four methods map to the four basic data actions, often called **CRUD**: Create (POST), Read (GET), Update (PUT), and Delete (DELETE). Here is a complete create call with every part labeled:

HTTP REQUEST · CREATE AN EMPLOYEE

```

POST /OpsAPI_B2WTechSupport/Employee HTTP/1.1           # 1 address + 2 method
Host: b2w-eus10.b2w.trimble.com
Authorization: Bearer <AccessToken>                        # 3 headers
Content-Type: application/json

{                                                           # 4 body
  "EmployeeID": "90001",
  "LastName": "Employee",
  "BusinessUnitUniqueName": "Organization"
}

```

The response always carries a status code, such as **201 Created** or **401 Unauthorized**, and plain text shaped as JSON. You will see the same responses in Postman, in Swagger, and in integration logs. [Question 19](#) explains every code.

3

How do I find a customer's Ops API URL?

SHORT ANSWER

Every cloud API URL follows one formula: `https://{cluster}/OpsAPI_{environment}` .
 Take the customer's Ops site address and put `OpsAPI_` in front of the environment name. For B2WTechSupport, that gives `https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport` .

`https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport`
CLUSTER (HOST)PREFIXENVIRONMENT

`/doc/index.html`
SWAGGER PAGE (OPTIONAL)

- The **cluster** is the server group the customer is hosted on, such as `b2w-eus10.b2w.trimble.com` .
- The **environment** is the customer's site name, such as `B2WTechSupport` .
- The customer's Ops website is `https://{cluster}/{environment}` . Their API is the same address with `OpsAPI_` added before the environment.
- The path is not case-sensitive. `opsapi_b2wtechsupport` works the same as `OpsAPI_B2WTechSupport` .

FOR

B2WTECHSUPPORT

ADDRESS

Ops website

`https://b2w-eus10.b2w.trimble.com/B2WTechSupport`

Ops API (base URL)

`https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport`

API docs (Swagger)

`https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport/doc/index.html`

Look it up in the Cloud Management Portal

If you have access to the Cloud Management Portal, open the customer's environment and find the **API Configuration** panel. The Ops API URL is listed under **Ops API Configuration**.

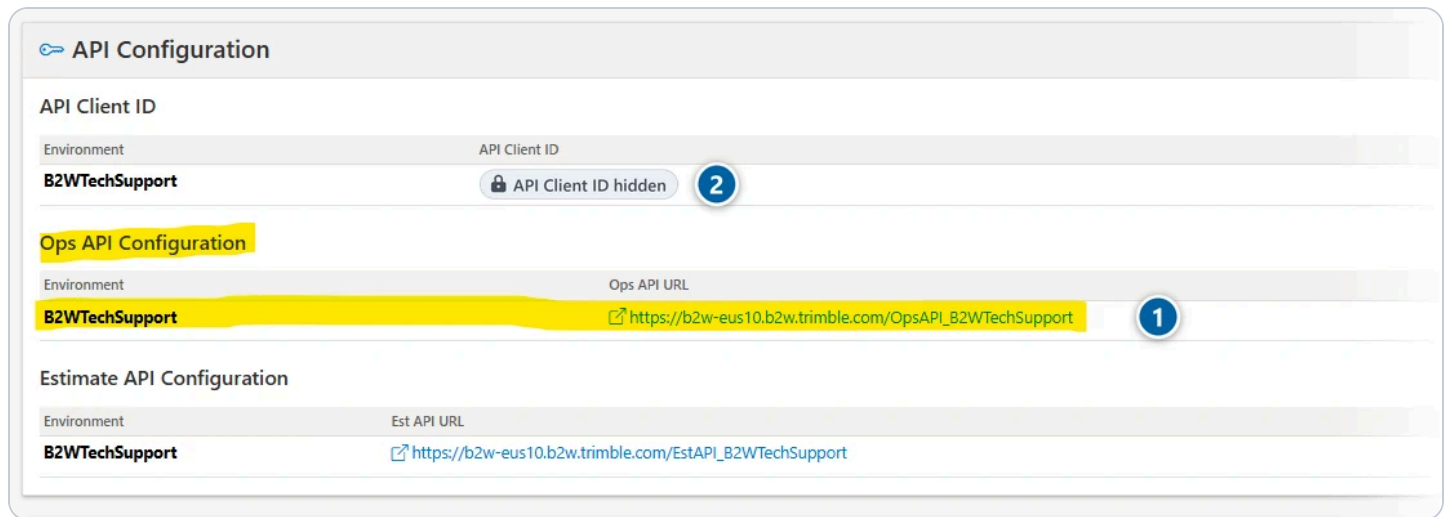


Figure 1. The API Configuration panel in the Cloud Management Portal. ① The Ops API URL for the environment. ② The environment's API Client ID, hidden here because it is a credential.



This panel shows credentials

The same panel lists the environment's API Client ID and, further right on the Ops API row, a **Copy Secret** button. Together they log in as the built-in System Administrator. Never paste them into an email, a chat, or a ticket.



Don't mix up the APIs

EstAPI_ (Estimate API) and MRAPI_ addresses on the same panel are different APIs with their own secrets. This guide covers the Ops API, OpsAPI_ , only.

TOOL

Build a customer's API URLs

Paste any address from their site, or type the parts

Paste a B2W address (optional)

`https://b2w-eus10.b2w.trimble.com/B2WTechSupport/...`

Works with the Ops site, the API, or a Swagger link. The cluster and environment fill in below.

Cluster (host)

`b2w-eus10.b2w.trimble.com`

Environment

`B2WTechSupport`

Ops website	<code>https://b2w-eus10.b2w.trimble.com/B2WTechSupport</code>
Ops API (baseUrl)	<code>https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport</code>
API docs (Swagger)	<code>https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport/doc/index.html</code>
Ping (no login)	<code>https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport/Ping/hello</code>
Version (no login)	<code>https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport/Version</code>
SystemInfo	<code>https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport/SystemInfo</code>

Open the Ping link to confirm the address. It answers without a login.



A typo answers with a 404 page

If the environment name is wrong, the server still responds, but with an HTML “404” error page instead of a result. If `/Ping/hello` returns that page, recheck the spelling and the `OpsAPI_` prefix.

4

Where is the API documentation?

SHORT ANSWER

Add `/doc/index.html` to the end of the Ops API URL. Each environment hosts its own Swagger catalog listing every endpoint, the fields it accepts, and the codes it returns.

`https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport/doc/index.html`

B2W Ops API v2 OAS3

../doc/v2/OpsAPI.json

General Information

The OpsAPI framework is a .NET Core based architecture to support one or more RESTful Web API endpoints to interact with the B2W Operational Suite.

All controllers in the framework conform to the .NET ApiController behavior and support HTTP verb-based endpoints augmented with attribute-based request routing.

Every endpoint is asynchronous in nature and returns an instance of IActionResult. Depending on the intent of an endpoint, it will either return just a status code, or a status code with relevant data.

No endpoint allows exceptions, if any, to "leak" to the .NET CLR. All exceptions which are propagated by the endpoints are centrally caught, reported, and handled. Exception information is logged to a log file, located in the **Logs** folder within the deployment folder.

The following sections provide useful information regarding configuration, and usage/testing guidelines. Click on the section title to expand/collapse that section. When you hover the mouse over a section, if the cursor changes to a pointer (a hand with a pointing-finger), you may expand/collapse that section.

Configuration

Usage Guidelines

API Documentation

The following sections provide detailed information regarding every functional area. Click on the section title to expand/collapse that section.

Figure 2. The top of the B2WTechSupport catalog. Click **Configuration** or **Usage Guidelines** to expand them. Usage Guidelines explains the login options in B2W's own words.

Scroll down to **API Documentation** to see every functional area. Click an area to expand its operations, then click an operation to see its parameters, example values, and response codes.

API Documentation

The following sections provide detailed information regarding every functional area. Click on the section title to expand/collapse that section.

BusinessUnit



Category



ConnectProjectInformation



Contact



Employee 1



DELETE /Employee Delete an entity.

GET /Employee Get a paginated collection of a model (mapped from an entity). 2

POST /Employee Create a new entity based on the specified model.

PUT /Employee Update an entity based on the specified model. First get the entity (based on its key), update relevant fields, and then PUT it.

GET /Employee/schema Get the JSON schema associated with a model.

EmployeeCertification



Figure 3. ① Each functional area expands into its operations. ② Each row is one operation. Employee has DELETE, GET, POST, PUT, and a /schema call.



Support takeaways from the training

- Always check Swagger for accurate field lists and the current schema. It reflects that customer's version.
- For an "is the API available?" ticket, your first action is to open the customer's Swagger page.

The same catalog is also available as a machine-readable OpenAPI 3 file at `/doc/v2/0psAPI.json`. Postman can import it to generate requests, but the team collection in [question 6](#) already has authentication set up, so start there.

Get set up

Install Postman, import the team collection, pick a login method, and find the credentials it needs.

5 How do I install Postman?

SHORT ANSWER

Download the free Postman desktop app from [postman.com/downloads](https://www.postman.com/downloads/) (<https://www.postman.com/downloads/>), install it, and sign in.

1 Download the app

Go to [postman.com/downloads](https://www.postman.com/downloads/) (<https://www.postman.com/downloads/>) and download the version for your computer (Windows, macOS, or Linux).

2 Install it

Run the installer. Postman opens when it finishes.

3 Sign in

Sign in, or create a free account. You need to be signed in to import and save collections.

 [Download Postman](#)

Postman is one tool among several

Postman sends requests and shows responses in a friendly window. Anything you do in it, you can also do from PowerShell or cURL. Where it helps, this guide shows all three.

How do I import the Postman collection?

SHORT ANSWER

Download the B2W Ops API collection from Google Drive, then import it into Postman with `Ctrl + O`. It appears as **B2W Ops API** in your sidebar, with a request for every area of the API.

1 Download the collection

Open the [collection on Google Drive](https://drive.google.com/file/d/1fjfhAF0z383mx1Io5qvFcjEHLsStrS9R/view?usp=sharing) (<https://drive.google.com/file/d/1fjfhAF0z383mx1Io5qvFcjEHLsStrS9R/view?usp=sharing>) and download the file. Sign in with your Trimble Google account if asked.

2 Open Import in Postman

Click **Import** at the top of the sidebar, or press `Ctrl + O` (`⌘ + O` on a Mac).

3 Add the file

Drag the downloaded file into the Import window, or click to browse for it.

4 Confirm

Finish the import. **B2W Ops API** appears under Collections.

 **Get the Postman collection**

What's inside

 **B2W Ops API** · collection variables live here

 **Auth** · the four ways to log in

 Login with user name and password

 Login with client ID and secret

 Login with TID uuid and API secret · used in the training

 LoginWithTID

 **Service**

 **BusinessUnit, Category, Contact, Employee, Equipment, Job ...** · one folder per functional area



From the training: "Auth" is a folder, not an endpoint

It only groups the login requests. Each one calls `GET /Login` (or `GET /LoginWithTID`) with a different pair of credential headers.

7

Which login method should I use?

SHORT ANSWER

Use a **TID UUID and User API Secret** when calls should act as a specific person. That keeps an audit trail and follows their security role. Use a **client ID and client secret** for unattended integrations, which run as the built-in System Administrator. Username and password still work but are being retired.

Every call except `/Ping` and `/Version` needs proof of identity. You prove it once, by sending credentials to the login endpoint **in HTTP headers**, never in the URL or the JSON body. The response gives you an access token to use on every other call.

METHOD	HEADERS TO SEND	CALLS RUN AS	USE IT FOR
TID UUID + User API Secret Used in the training	<code>tiduuid</code> <code>apiSecret</code>	That specific Ops user, limited by their security role	People testing the API, and integrations that need an audit trail of who changed what
Client ID + client secret	<code>clientId</code> <code>clientSecret</code>	The built-in System Administrator (elevated access)	Unattended integrations, such as scheduled overnight imports and middleware
Username + password Being retired	<code>userName</code> <code>password</code>	That Windows / Ops user	Older integrations. Plan to move them to TID or client credentials.

METHOD	HEADERS TO SEND	CALLS RUN AS	USE IT FOR
LoginWithTID Separate endpoint	A Trimble ID access token as the bearer token	The Ops user linked to that Trimble ID	Apps where the person already signed in with Trimble ID



From the training: client ID + secret vs. TID

A client ID and secret give system-administrator-level access. TID keeps an audit trail of what was changed and takes on the persona of the user, following the limits set in their security role.

The order /Login checks headers in

GET /Login looks for one complete pair of credentials, in this order, and uses the first pair it finds:

PRIORITY 1 · USER

- userName
- password

PRIORITY 2 · CLIENT

- clientId
- clientSecret

PRIORITY 3 · API

- tiduuid
- apiSecret



If no valid pair is in the headers, GET /Login returns 400 with {"Error": "Authentication information must be specified in the request header"}. That is the exact response B2WTechSupport gives today.

Username and password details

The `userName` header must match the Windows account name stored on the Ops user, written as `DOMAIN\user` or `user@domain`. Ops checks the format, verifies the password with the Windows domain controller, then confirms the account exists in the Ops user table. The token it issues is tied to that person, so every action is traceable to them.

LoginWithTID

This is a different endpoint, `GET /LoginWithTID`. Instead of a credential pair, you send a Trimble ID access token that the person got by signing in to Trimble ID. Ops exchanges it for an Ops access token.

INPUT

Trimble ID bearer token

The caller's existing Trimble ID token.



OPS VALIDATION

Checks signature and lifetime

Pulls out the user ID and email.



OUTPUT

Ops access token

Linked to the matching Ops person.



LoginWithTID returns 404? Fix the link, not the password.

A 404 here means Trimble accepted the person, but no Ops user matches their Trimble ID or mobile email address. Fix the Ops user record (its **TID / Mobile E-mail Address**). Do not reset the password; the Trimble sign-in is already valid.



Trimble Identity (TID) mode

Client ID/secret and Trimble ID logins work when the site runs in Trimble Identity mode. To check a site, open `/SystemInfo`. B2WTechSupport returns `"IsTIDEnabled": true`.



No login method bypasses API security

Whichever way you sign in, every call is still checked against the API permissions and Ops validation rules.

8

How do I find my TID UUID and User API Secret?

SHORT ANSWER

In B2W Ops, open your user record and click **User API Secret**. The dialog shows your **TID ID**, which is the `tiduuid`, and your **User API Secret**, which is the `apiSecret`.

- 1 Open Users**
From the gear (⚙️) menu at the top right of B2W Ops, open **Users**, then open the person's record. You land on **User View**.
- 2 Check the TID / Mobile E-mail Address**
It must be filled in. This is what links the Ops user to their Trimble ID.
- 3 Click User API Secret**
It sits under the e-mail address, between **Show TID ID** and **Manage TID**.
- 4 Note the security role**
Calls made with these credentials can only do what this role allows.

B2W

HOME JOBS SCHEDULE TRACK MAINTAIN

Users 1

User View - Dartagnan Birnie

View User List Edit Delete Create New User

General Information

First Name	Dartagnan
Business Unit	Organization
Inactive	No
Title	
TID / Mobile E-mail Address 2	dartagnan_birnie@trimble.com 3
	Show TID ID User API Secret Manage TID
Licenses	Track (Full Access), Field Employee (Full Access), Schedule (Full Access), Maintain Manager (Full Access)
Notes	

Security Roles

Name ▲
System Administrator - All Access 4

Figure 4. User View in B2WTechSupport. The numbers match the steps above.

The **User API Secret** dialog opens:

- 1 TID ID**
This is the `tiduuid`. Put it in the `tidUuid` variable in Postman.
- 2 User API Secret**
This is the `apiSecret`. Put it in the `apiSecret` variable.
- 3 Copy to Clipboard**
Copies the secret, which is masked on screen.
- 4 Generate User API Secret**
Creates a secret when the user doesn't have one. A value must be generated before anyone can call the API as this user.

User API Secret

A User API Secret Value can be used, in addition with this user's TID ID, to allow access to API calls as this user. A value must be generated for this user before any API access can be done as this specific user.

TID ID	TID ID hidden	1
User API Secret		2

Copy to Clipboard

3

Generate User API Secret

4

Close

Figure 5. The User API Secret dialog. The TID ID is hidden in this guide.



Treat "Generate" as a reset

If an integration already uses this user's secret, coordinate with its owner before generating a new one.



Same values, different spellings

The training notes call this pair "tidUID and apiSecret". In the login headers they are `tiduuid` and `apiSecret`. In the Postman collection they are the `tidUuid` and `apiSecret` variables.

How do I find my clientID and clientSecret?

SHORT ANSWER

In B2W Ops, go to **Integrations & Add-ins** and click **Manage API Key**. That dialog holds the client ID and client secret for the environment.

1 Open Integrations & Add-ins

From the gear (⚙️) menu at the top right of B2W Ops, open **Integrations & Add-ins**.

2 Click Manage API Key

It is the first button at the top right of the page. The dialog shows the client ID and client secret that B2W Ops generated for this environment.

3 Use them in Postman

Put them in the `clientId` and `clientSecret` variables, then send **Auth** → **Login with client ID and secret**.

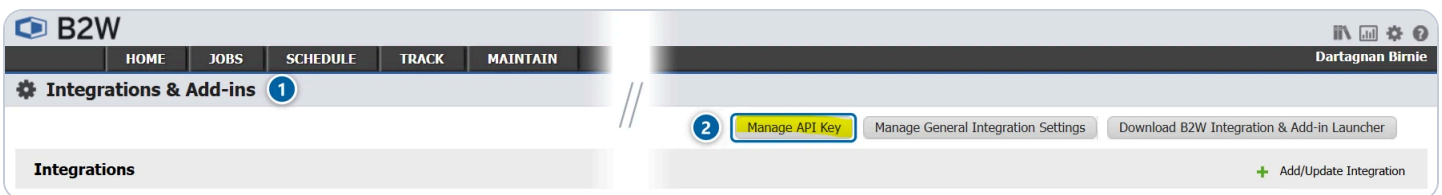


Figure 6. ① The Integrations & Add-ins page. ② Manage API Key. The middle of the page is trimmed to save space.



Never send a client ID and secret over email

These credentials log in as the built-in System Administrator, with full access to the customer's data. Share them only through a secure channel, and only with people who need them.



Don't replace a key that is in use

Ops validates logins against the key stored in its settings. If a new key is generated, integrations still using the old secret start failing with **401** until they are updated.

When to use it: scheduled overnight jobs and middleware that shouldn't depend on one person's credentials. Two responses are specific to this method:

- **401** The client ID or secret doesn't match the values stored in Ops system settings.
- **404** The credentials are valid, but the built-in System Administrator record is missing in Ops.

Support staff with Cloud Management Portal access can also find the environment's API Client ID and Ops API secret on the API Configuration panel ([Figure 1](#)).

10 How do I fill out the Postman variables?

SHORT ANSWER

Click the **B2W Ops API** collection itself (the top level), open the **Variables** tab, and fill in `baseUrl` plus the credential pair you are using. Save with `Ctrl + S`.

1 Open the collection's Variables tab

In the sidebar, click **B2W Ops API** at the very top, then the **Variables** tab.

2 baseUrl

The Ops API URL with no slash at the end, for example `https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport`.

3 tidUuid

The TID ID from the User API Secret dialog.

4 apiSecret

The User API Secret.

5 accessToken

Leave it. It holds your token after you log in ([question 11](#)).

B2W Ops API Run

Overview
Authorization
Scripts
Tests
Variables
Runs

Variable	Value
baseUrl	https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport
accessToken	Filled in after you log in
userName	
password	
clientId	
clientSecret	
tidUuid	Your TID ID (hidden)
apiSecret	Your User API Secret (hidden)
trimbleToken	
objectId	
pageSize	100
pingInput	hello
trueUpBatchNumber	
categoryType	
jobNumber	
maintenanceRequestId	
Add variable	

Figure 7. The collection’s Variables tab, set up for B2WTechSupport. Your own TID ID and secret are hidden here.

VARIABLE	WHAT GOES IN IT	EXAMPLE OR DEFAULT
baseUrl	The customer’s Ops API URL	https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport
accessToken	The AccessToken from the login response	Filled in after you log in
tidUuid · apiSecret	TID ID and User API Secret (question 8)	Your own values

VARIABLE	WHAT GOES IN IT	EXAMPLE OR DEFAULT
<code>clientId</code> · <code>clientSecret</code>	From Manage API Key (question 9)	The environment's values
<code>userName</code> · <code>password</code>	Windows account credentials (being retired)	<code>DOMAIN\user</code>
<code>trimbleToken</code>	A Trimble ID access token, for LoginWithTID	Optional
<code>objectId</code>	The <code>objectId</code> of a record to update or delete	<code>a4066d42-4027-4d5d-87a0-a30600f8d4cf</code>
<code>pageSize</code>	Records per page	<code>100</code>
<code>pingInput</code>	Text that <code>/Ping</code> echoes back	<code>hello</code>
<code>trueUpBatchNumber</code> · <code>categoryType</code> · <code>jobNumber</code> · <code>maintenanceRequestId</code>	Inputs for specific requests in those folders	As needed



Treat secrets like passwords

Don't share or export the collection while `apiSecret` , `clientSecret` , `password` , or `accessToken` hold real values. Clear them first.

Make calls

Get a token, use it, check connectivity, query data, and make changes safely.

11 How do I generate a Bearer Token?

SHORT ANSWER

Send `GET {baseUrl}/Login` with your credentials in the headers. The `AccessToken` in the JSON response is your bearer token. In Postman: **Auth** → **Login with TID uuid and API secret** → **Send**.

1 Open the Auth folder

In the B2W Ops API collection.

2 Select “Login with TID uuid and API secret”

It is a `GET` request to `{{baseUrl}}/Login`.

3 Check the headers

On the **Headers** tab, `tiduuid` and `apiSecret` pull their values from your variables.

4 Click Send

Or press `Ctrl + Enter`.

5 Look for 200 OK

Any other code means the login failed. See [question 20](#).

6 Find the AccessToken

That value is your bearer token. Make sure it is in the `accessToken` variable before you call other endpoints.

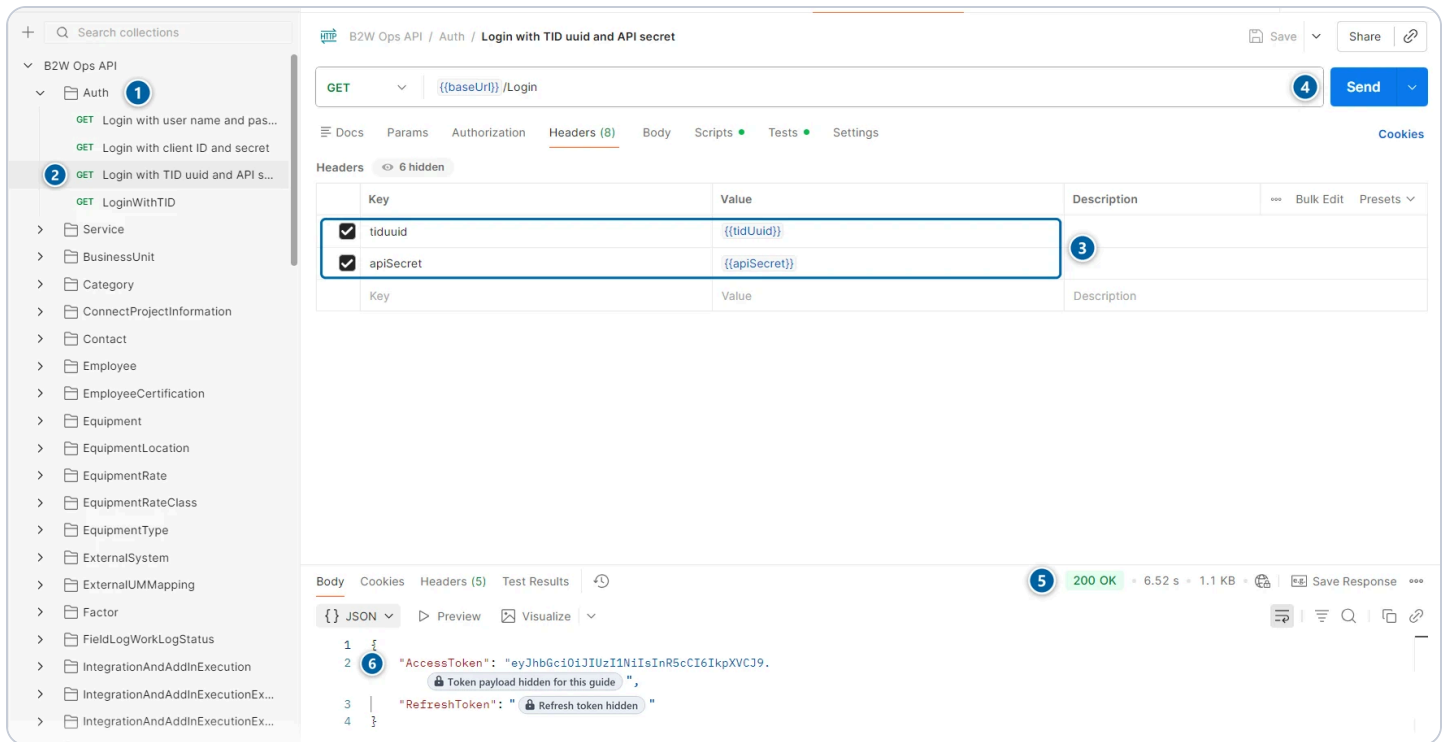


Figure 8. Generating a token for B2WTechSupport. The numbers match the steps above. The token values are hidden.

The same request outside Postman

POSTMAN

GET <code>{{baseUri}}/Login</code>	
HEADERS	
tiduuid	<code>{{tidUuid}}</code>
apiSecret	<code>{{apiSecret}}</code>
BODY	
none	Login never takes a body.

POWERSHELL

POWERSHELL

```
$baseUrl = "https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport"

# Credentials go in the headers, never in the URL or the body
$login = Invoke-RestMethod -Method Get -Uri "$baseUrl/Login" -Headers @{
    tiduuid    = "<your TID ID>"
    apiSecret  = "<your User API Secret>"
}

$token = $login.AccessToken # this is your bearer token
```

CURL

CURL · BASH, MACOS, GIT BASH

```
curl -s 'https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport/Login' \
-H 'tiduuid: <your TID ID>' \
-H 'apiSecret: <your User API Secret>'
```

RAW HTTP

HTTP REQUEST

```
GET /OpsAPI_B2WTechSupport/Login HTTP/1.1
Host: b2w-eus10.b2w.trimble.com
tiduuid: <your TID ID>
apiSecret: <your User API Secret>
```

A successful login returns **200 OK** and a small JSON object:

RESPONSE · 200 OK

```
{
  "AccessToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOi...",
  "RefreshToken": "<refresh token>"
}
```

Token facts

- **AccessToken and bearer token mean the same thing.** It is a signed JWT.
- A token lasts **1 day** by default. The site setting `Security:JWT:Lifetime` controls this.

- Per the training, a token also stops working once you generate a new one, so always use your latest.
- When a token expires, data calls return **401 Unauthorized** . Log in again for a new one.
- The response also includes a `RefreshToken` . The catalog has no refresh endpoint, so when your token expires, simply log in again.



A token is a password for a day

Anyone holding it can act as you until it expires. Don't paste tokens into tickets, chats, or screenshots.

12 How do I use the token on a data call?

SHORT ANSWER

Send it in the `Authorization` header as the word `Bearer` , one space, then the token. Add `Content-Type: application/json` when the call has a body.

STEP 1 · LOGIN

GET /Login

Passwords, client secrets, and Trimble tokens go here only.



STEP 2 · TOKEN

AccessToken (signed JWT)

Lasts 1 day by default.



STEP 3 · DATA CALLS

Authorization: Bearer <AccessToken>

Required on every other call.

Signing in and calling data are separate steps. Your secrets only ever go to the login endpoint; every other call carries the token instead. Here is a first data call that reads five employees:

POSTMAN

GET `{{baseUrl}}/Employee?$top=5`

AUTHORIZATION

Type	Bearer Token, inherited from the collection
Token	<code>{{accessToken}}</code>

POWERSHELL

POWERSHELL · CONTINUES FROM QUESTION 11

```
$headers = @{ Authorization = "Bearer $token" }  
  
# For GET, -Body becomes the query string: /Employee?$top=5  
$employees = Invoke-RestMethod -Method Get -Uri "$baseUrl/Employee" -Headers  
$headers -Body @{ '$top' = 5 }  
$employees | Select-Object EmployeeID, FirstName, LastName
```

CURL

CURL · BASH, MACOS, GIT BASH

```
curl -s -G 'https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport/Employee' \  
--data-urlencode '$top=5' \  
-H 'Authorization: Bearer <AccessToken>'
```

RAW HTTP

HTTP REQUEST

```
GET /OpsAPI_B2WTechSupport/Employee?$top=5 HTTP/1.1  
Host: b2w-eus10.b2w.trimble.com  
Authorization: Bearer <AccessToken>
```

The response is a JSON array of employee records. This sample comes from the catalog's own example:

RESPONSE · 200 OK (SHORTENED)

```
[
  {
    "ObjectID": "a4066d42-4027-4d5d-87a0-a30600f8d4cf",
    "EmployeeID": "12865",
    "FirstName": "RaLph",
    "LastName": "Newman",
    "JobTitle": "Field Employee",
    "IsFieldEmployee": true,
    "IsInactive": false,
    "BusinessUnitUniqueName": "Northern Division\\Paving",
    "RowVersion": 1
  }
]
```



In the collection, auth is already wired up

The collection's **Authorization** tab sends `{{accessToken}}` as a bearer token, and its requests inherit that setting. If a call returns `401` right after you logged in, check that `accessToken` holds your newest token and that the request's Authorization tab is set to inherit from its parent.



Get the header exactly right

`Authorization: Bearer eyJhbGciOi...` uses the word `Bearer`, a single space, and the token with no quotes around it.

13

How do I check connectivity without logging in?

SHORT ANSWER

Open `{baseUrl}/Ping/hello` in a browser. If you get the server name back, the API is reachable. `/Version` returns the API version. Neither one needs a login.

B2WTECHSUPPORT RETURNED (OCT 2, 2026)		
CALL		WHAT IT TELLS YOU
GET /Ping/hello	(b2w-eus10-iis2) hello	The site is up. It echoes your text with the name of the server that answered.
GET /Version	26.1.2.1	The Ops API build the customer is running.
GET /SystemInfo	{"IsTIDEnabled":true,"IsEMSLicensed":false}	Whether the site runs in Trimble Identity mode. Also answered without a login on B2WTechSupport.

Reading the result

YOU SEE	IT MEANS
The server name and your text	The site and network are healthy. Move on to the login.
A timeout or “connection refused”	The API site, the IIS web server, or the network is down. Check connectivity, routing, and the IIS host.
An HTML “404” page	The server is up but that address doesn’t exist. Check the environment name and the <code>OpsAPI_</code> prefix.
401 on <code>/Employee</code> or <code>/Employee/schema</code>	Expected without a token. Every data call, including <code>/schema</code> , needs one.

In the Postman collection, the Ping request uses the `pingInput` variable, which defaults to `hello`.

How do I filter, sort, and page through results?

SHORT ANSWER

GET endpoints accept OData query options: `$filter`, `$select`, `$orderby`, `$top`, and `$skip`. Each call returns at most 100 records, so use `$skip` to fetch the next page.

OPTION	WHAT IT DOES	EXAMPLE FROM THE CATALOG
<code>\$filter</code>	Returns only matching records	<code>\$filter=EmployeeID eq '12865'</code>
<code>\$select</code>	Returns only the fields you list	<code>\$select=EmployeeID</code>
<code>\$orderby</code>	Sorts the results	<code>\$orderby=EmployeeID, LastName, FirstName asc</code>
<code>\$top</code>	Limits how many records come back (100 at most)	<code>\$top=1</code>
<code>\$skip</code>	Skips records, for paging	<code>\$skip=1</code>

GET `/Employee` Get a paginated collection of a model (mapped from an entity).

Parameters

Name	Description
queryOptions	OData query options
object (query)	<pre> \$orderby=EmployeeID, LastName, FirstName asc \$select=EmployeeID \$skip=1&\$top=1 \$filter=EmployeeID eq '12865' The query options may be combined to create compound queries. </pre>

Figure 9. Every GET operation in Swagger lists the OData options it accepts. They can be combined into one query.

Examples for B2WTechSupport

POSTMAN URLS

One employee by ID

```
{{baseUrl}}/Employee?$filter=EmployeeID eq '12865'
```

Just three fields, sorted by last name

```
{{baseUrl}}/Employee?$select=EmployeeID,FirstName,LastName&$orderby=LastName
```

Active jobs, page 2 (records 101-200)

```
{{baseUrl}}/Job?$filter=IsActive eq true&$orderby=JobNumber&$top=100&$skip=100
```

Paging through more than 100 records

PAGE	ADD TO THE URL	RECORDS
1	<code>\$top=100&\$skip=0</code>	1–100
2	<code>\$top=100&\$skip=100</code>	101–200
3	<code>\$top=100&\$skip=200</code>	201–300

Keep going until a page comes back with fewer than 100 records. Sort with `$orderby` so the pages stay in a stable order.



Exactly 100 rows usually means “more to come”

A GET that returns exactly 100 records has hit the page size, not the end of the data. Ask for the next page before assuming records are missing.



OData syntax tips

- Put text values in single quotes: `LastName eq 'Newman'` .
- Compare with `eq` , `ne` , `gt` , `ge` , `lt` , and `le` . Combine conditions with `and` and `or` .
- Join options with `&` . Postman and browsers encode the spaces for you.

How do I create, update, or delete a record?

SHORT ANSWER

POST creates, **PUT** updates, and **DELETE** removes. Before you write anything, GET a record from the same area so you can copy its exact shape.



Practice in B2WTechSupport only

Write calls change real data. Test them in the B2WTechSupport environment, and never in a customer's environment unless the customer has asked for the change.

Create a record (POST)

Send the new record as JSON. For an employee, the catalog requires `EmployeeID`, `LastName`, and `BusinessUnitUniqueName`. A success returns **201 Created**; a body that isn't a valid record returns **422**.

POSTMAN

POST `{{baseUrl}}/Employee`

BODY · RAW · JSON

EmployeeID	"90001"
FirstName	"Test"
LastName	"Employee"
BusinessUnitUniqueName	"Organization"
IsFieldEmployee	true

POWERSHELL

POWERSHELL

```
$body = @{"EmployeeID" = "90001"
"FirstName" = "Test"
"LastName" = "Employee"
"BusinessUnitUniqueName" = "Organization"
"IsFieldEmployee" = $true
} | ConvertTo-Json

Invoke-RestMethod -Method Post -Uri "$baseUrl/Employee" -Headers $headers -Body
$body -ContentType "application/json"
```

CURL

CURL · BASH, MACOS, GIT BASH

```
curl -s -X POST 'https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport/Employee' \
-H 'Authorization: Bearer <AccessToken>' \
-H 'Content-Type: application/json' \
-d
'{"EmployeeID":"90001","FirstName":"Test","LastName":"Employee","BusinessUnitUniqueName":"Organization","IsFieldEmployee":true}'
```

RAW HTTP

HTTP REQUEST

```
POST /OpsAPI_B2WTechSupport/Employee HTTP/1.1
Host: b2w-eus10.b2w.trimble.com
Authorization: Bearer <AccessToken>
Content-Type: application/json

{
  "EmployeeID": "90001",
  "FirstName": "Test",
  "LastName": "Employee",
  "BusinessUnitUniqueName": "Organization",
  "IsFieldEmployee": true
}
```

Update a record (PUT)

The catalog's rule: first GET the record by its key, change the fields you need, then PUT the whole record back. Keep its `ObjectID` and `RowVersion` exactly as you received them. A success returns `200 OK`.

POWERSHELL · GET, CHANGE, PUT

```
# 1. GET the current record
$record = Invoke-RestMethod -Uri "$baseUrl/Employee" -Headers $headers -Body @{'$filter' = "EmployeeID eq '90001'" } | Select-Object -First 1

# 2. Change only what you need
$record.JobTitle = "Field Employee"

# 3. PUT the whole record back, ObjectID and RowVersion included
Invoke-RestMethod -Method Put -Uri "$baseUrl/Employee" -Headers $headers -Body ($record | ConvertTo-Json -Depth 5) -ContentType "application/json"
```

If a PUT returns `409 Conflict`, the update clashes with the record's current state, for example because someone changed it after you read it. GET it again and reapply your change.

Delete a record (DELETE)

Send the record's `ObjectID` as a query parameter or as a header named `ObjectID`. A success returns `204 No Content`.

HTTP REQUEST

```
DELETE /OpsAPI_B2WTechSupport/Employee?ObjectID=<ObjectID> HTTP/1.1
Host: b2w-eus10.b2w.trimble.com
Authorization: Bearer <AccessToken>
```

Rules to remember

- **You cannot change an ObjectID through the API.** It is the record's permanent key.
- Each action needs the matching API permission in the caller's security role: Create, View, Edit, or Delete ([question 20](#)).
- A read-only license cannot make changes, even if the security role would allow them.

From the training: build your POST from a GET

If you want to put data into Ops, run a GET in that area first. It shows you the full outline of a record, and you can build your POST around it. `GET /{Area}/schema` returns the formal JSON schema too (it needs a token).

16 Which endpoints are available?

SHORT ANSWER

The B2WTechSupport catalog (version 26.1.2.1) lists 63 functional areas and 272 operations. Most areas support GET, POST, PUT, and DELETE plus a `/schema` call. Search the list below, then open Swagger for the fields.

TOOL Endpoint explorer

Showing 64 of 64 endpoint groups · 272 operations in all

/Login

GET

Credentials in headers → AccessToken

/LoginWithTID

GET

Trimble ID token → Ops AccessToken

/Ping

GET /Ping/{input}

No login needed

/Version

GET

No login needed

/SystemInfo

GET

Trimble ID mode and licensing

/Job

GET POST PUT DELETE + /schema

/JobChangeOrder

GET POST PUT DELETE + /schema

/JobCostBreakdownElement

GET POST PUT DELETE + /schema

/JobCostBreakdownElementMaterial

GET POST PUT DELETE + /schema

/JobEstimateItem

GET POST PUT DELETE + /schema

/JobFileAttachment

GET **POST** **PUT** **DELETE** + /schema

/JobLaborRateClass

GET **POST** **DELETE** + /schema

/JobMaterial

GET **POST** **PUT** **DELETE** + /schema

/JobOrganization

GET **POST** **DELETE** + /schema

/JobOverheadAccount

GET **POST** **PUT** **DELETE** + /schema

/JobProductionAccount

GET **POST** **PUT** **DELETE** + /schema

/JobProductionTarget

GET **POST** **PUT** **DELETE** + /schema

/JobProjectManager

GET **POST** **DELETE** + /schema

/JobSite

GET **POST** **PUT** **DELETE** + /schema

/JobSubcontractorQuote

GET **POST** **PUT** **DELETE** + /schema

/JobTnMWorkItem

GET **POST** **PUT** **DELETE** + /schema

/ConnectProjectInformation

GET **POST** **DELETE** + /schema

/BusinessUnit

GET + /schema

/Contact

GET **POST** **PUT** **DELETE** + /schema

/Employee

GET **POST** **PUT** **DELETE** + /schema

/EmployeeCertification

GET **POST** **PUT** **DELETE** + /schema

/Organization

GET **POST** **PUT** **DELETE** + /schema

/User

GET **POST** **PUT** **DELETE** + /schema

/LaborType

GET **POST** **PUT** **DELETE** + /schema

/LaborTypeRate

GET **POST** **PUT** **DELETE** + /schema

/LaborRateClass

GET **POST** **PUT** **DELETE** + /schema

/Equipment

GET **POST** **PUT** **DELETE** + /schema

/EquipmentLocation

GET + /schema

/EquipmentRate

GET **POST** **PUT** **DELETE** + /schema

/EquipmentRateClass

GET **POST** **PUT** **DELETE** + /schema

/EquipmentType

GET **POST** **PUT** **DELETE** + /schema

/InternalTruck

GET **POST** **PUT** **DELETE** + /schema

/SubcontractedTruck

GET **POST** **PUT** **DELETE** + /schema

/MaintenanceRequest

GET **POST** **PUT** **DELETE** + /schema

GET /MaintenanceRequest/enums

/MaintenanceRequestAttachment

GET **POST** **PUT** **DELETE** + /schema

GET

/MaintenanceRequestAttachment/withBytes

/MaintenanceRequestComment

GET **POST** **PUT** **DELETE** + /schema

/Part

GET **POST** **PUT** **DELETE** + /schema

/PartInventory

GET **POST** **PUT** **DELETE** + /schema

/TrackWorkOrder

GET **POST** **PUT** **DELETE** + /schema

/Category

GET **POST** **PUT** **DELETE** + /schema

/Subcategory

GET **POST** **PUT** **DELETE** + /schema

/Factor

GET **POST** **PUT** **DELETE** + /schema

/Material

GET **POST** **PUT** **DELETE** + /schema

/Miscellaneous

GET **POST** **PUT** **DELETE** + /schema

/OverheadAccount

GET **POST** **PUT** **DELETE** + /schema

/ProductionAccount

GET **POST** **PUT** **DELETE** + /schema

/Place

GET **POST** **PUT** **DELETE** + /schema

/UnitOfMeasure

GET + /schema

/ExternalSystem

GET **POST** **PUT** **DELETE** + /schema

/ExternalUMMapping

GET **POST** **PUT** **DELETE** + /schema

/ResourceEvent

GET **POST** **PUT** **DELETE** + /schema

/ProductionCrewTemplate

/TransportCrewTemplate

GET + /schema	GET + /schema
/FieldLogWorkLogStatus GET PUT + /schema	/IntegrationAndAddInExecution GET + /schema
/IntegrationAndAddInExecutionExportItem GET + /schema	/IntegrationAndAddInExecutionExportItemRecord GET + /schema
/TrueUpMaterialBatch GET POST DELETE GET /TrueUpMaterialBatch/{TrueUpBatchNumber}/TrueUpMaterial GET /TrueUpMaterialBatch/{TrueUpBatchNumber}/TrueUpAccountMaterial	/TrueUpPriceAndCostBatch GET POST DELETE GET /TrueUpPriceAndCostBatch/{TrueUpBatchNumber}/TrueUpAccount GET /TrueUpPriceAndCostBatch/{TrueUpBatchNumber}/TrueUpEstimateItem
GET read POST create PUT update DELETE remove /schema field definitions	

The list reflects the B2WTechSupport catalog on October 2, 2026. Environments on other versions may differ, and their own Swagger page is always the source of truth.

Troubleshoot

A repeatable way to work Ops API tickets, an interactive troubleshooter, and fixes for the errors you will see most.

17 How do I work an Ops API ticket?

SHORT ANSWER

Ask the five core diagnostic questions in order. They separate the API surface, the connection, the login, and the data call, so you find the failing step quickly.

01

API surface

Is it the Ops API (JSON) or the older Web Services (SOAP XML)? Are they using the API or Web Services?

02

Base URL and connectivity

Is the base URL right, and does `GET /Ping/hello` succeed? Always check connectivity first.

03

Authentication method

Which sign-in method do they use: user (username and password), TID, or client ID and secret?

04

Login response, on its own

Can they log in and get a token back? Look at the login's status code and body separately from the data call's.

05

Header authorization

Does the data call carry the correct `AccessToken` in an `Authorization: Bearer` header?

Collect these details

- ✓ Customer and environment name
- ✓ Endpoint and method, such as `GET /Job`
- ✓ Status code and response body, with secrets removed
- ✓ The tool or integration making the call
- ✓ The Ops API URL they call
- ✓ Login method they use
- ✓ Date, time, and time zone of the failure
- ✓ Whether it ever worked, and what changed



Ask for evidence, not secrets

Request screenshots or logs of the request and response with credentials and tokens hidden. Never ask a customer to email a client ID and secret.

18

Where is this ticket stuck?

SHORT ANSWER

Answer a few questions about what the customer sees. The troubleshooter walks the five diagnostic questions and lands on the likely cause and fix.

19

What do the status codes mean?

SHORT ANSWER

2xx means it worked. 4xx means the request was wrong: bad input, a bad or missing token, missing permissions, or something not found. 5xx means Ops hit an unexpected error. The same code can mean different things on the login call and on a data call.

CODE	MEANING	ON <code>/LOGIN</code>	ON A DATA CALL
<code>200 OK</code>	Success	Token issued	Data returned or record updated

CODE	MEANING	ON /LOGIN	ON A DATA CALL
201 Created	Record created	–	A POST succeeded
204 No Content	Done, nothing to return	–	A DELETE succeeded
400 Bad Request	Incomplete or malformed request	No credentials in the headers, or the user name is invalid	Missing parameters or a malformed query
401 Unauthorized	Unsigned caller or bad token	Credentials rejected	Bearer token missing, invalid, or expired
403 Forbidden	Not enough privileges	–	The security role lacks that API permission
404 Not Found	Resource or user missing	No matching Ops user, or the System Administrator record is missing	Record not found. An HTML 404 page means a wrong URL.
406 Not Acceptable	Unexpected format	The System Administrator's Windows account name isn't in the expected format	–
409 Conflict	Conflicting update	–	A PUT clashes with the record's current state
422 Unprocessable	Body isn't a valid record	–	A POST or PUT body is missing required fields or is malformed
500 Server Error	Unexpected internal Ops error	Capture the request, response, and time, then escalate	

SHORT ANSWER

Find the symptom below. Each card gives the likely cause and the fix. Isolate the failing step first: connection, login, or data call.

403 Forbidden: “insufficient privileges”

A 403 almost always means the user’s Ops security role doesn’t grant the API permission the call needs. The response names the missing privilege:

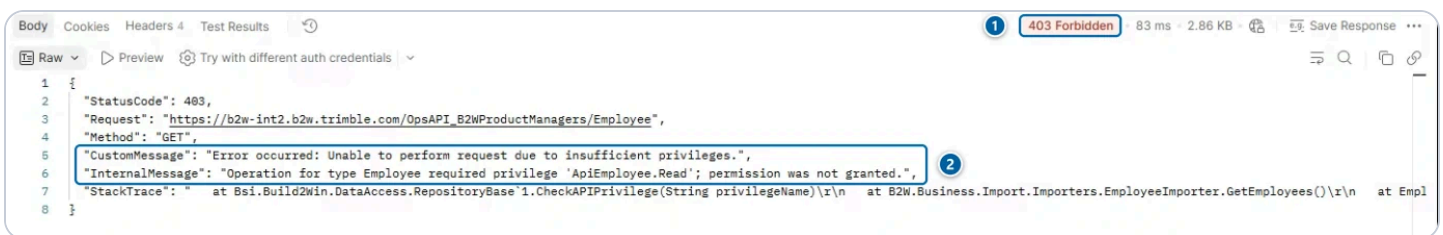


Figure 10. ① The 403 status. ② The messages explain why: the call needed `ApiEmployee.Read`, and the user’s role doesn’t grant it.

- 1 Open the user’s security role in Ops**
From their User View (Figure 4), click the role name.
- 2 Find the API section and the area’s row**
Check the column for the action: **View** for GET, **Create** for POST, **Edit** for PUT, **Delete** for DELETE. A filled green dot means granted.

API

1

Description	Create	View	Edit	Delete
Account	☐	☒	☐	☐
Business Unit		☒		
Category	☐	☒	☐	☐
Crew Template	☐	☒	☐	☐
Employee 2	☐	☒	☐	☐
Equipment	☐	☒	☐	☐
Equipment Rate Class	☐	☒	☐	☐
Equipment Type	☐	☒	☐	☐
External System	☐	☒	☐	☐
External UM Mapping	☐	☒	☐	☐
Factor	☐	☒	☐	☐
Field Log\Work Log		☒	☐	
Integration\Add-in		☒	☐	
Job	☐	☒	☐	☐
Labor Rate Class	☐	☒	☐	☐
Labor Type	☐	☒	☐	☐
Material	☐	☒	☐	☐
Miscellaneous	☐	☒	☐	☐
Organization	☐	☒	☐	☐
Organization Contact	☐	☒	☐	☐
Part	☐	☒	☐	☐
PartInventory	☐	☒	☐	☐
Place	☐	☒	☐	☐
Resource Event	☐	☒	☐	☐
Track Work Order	☐	☒	☐	☐
Truck	☐	☒	☐	☐
True-up	☐	☒		☐
Unit Of Measure		☒		
User	☐	☒	☐	☐

Figure 11. ① The API section of a security role. ② With this role, `GET /Employee` works (View granted), but `POST /Employee` returns 403 (Create not granted).



Licenses matter too

A read-only license cannot make changes in Ops, even if the security role allows them. Per the training, it cannot even run a GET on employees.

Every other common error

No response Ping fails or the connection is refused

LIKELY CAUSE

The API site, the IIS web server, or the network is down entirely.

FIX

Check network connectivity, routing, and the IIS host status. Escalate if the site itself is down.

404 page Ping returns an HTML "404" page

LIKELY CAUSE

The address is wrong: a typo in the environment name, or a missing `OpsAPI_` prefix.

FIX

Rebuild the URL with the formula in [question 3](#) and try `/Ping/hello` again.

400 Login: "Authentication information must be specified in the request header"

LIKELY CAUSE

No complete credential pair in the headers. The values may be in the URL or body, or a header name is misspelled.

FIX

Send exactly one pair as headers: `tiduuid` + `apiSecret`, `clientId` + `clientSecret`, or `userName` + `password`.

401 Login rejected

LIKELY CAUSE

With a user name: Windows rejected the password. With a client ID: the ID or secret doesn't match Ops system settings. With a TID UUID: the TID ID or User API Secret is wrong.

FIX

Re-copy the values from their source ([question 8](#) or [question 9](#)). Ask whether anyone generated a new secret or key recently.

404 Login: user not found

LIKELY CAUSE

With a user name: the account isn't set up as an Ops user. With client credentials: the built-in System Administrator record is missing in Ops.

FIX

Confirm the user exists and is active in Ops. For the System Administrator record, escalate.

404 LoginWithTID: user linkage failure

LIKELY CAUSE

Trimble accepted and authenticated the person, but no Ops user matches their Trimble ID or mobile email address.

FIX

Fix the link on the Ops user record (TID / Mobile E-mail Address). Do not reset the password; the Trimble sign-in is valid.

401 Login works, then the data call returns 401

LIKELY CAUSE

The bearer token is missing from the `Authorization` header, or it has expired.

FIX

Check the header reads `Authorization: Bearer <AccessToken>`. Log in again for a fresh token; they last 1 day.

200 A GET returns exactly 100 rows

LIKELY CAUSE

The default page size, not missing data.

FIX

Page with `$top` and `$skip` until a page has fewer than 100 records ([question 14](#)).

XML The request or response is XML

LIKELY CAUSE

XML message bodies and `.asmx` URLs belong to the older SOAP Web Services. They don't map to Ops API 2.0 endpoints.

FIX

Handle it as a Web Services ticket. Where it makes sense, recommend moving the integration to the Ops API.

422 POST or PUT is rejected

LIKELY CAUSE

The body isn't a valid record: a required field is missing, a field name is wrong, or the JSON is malformed.

FIX

Compare the body with a GET of an existing record or with `/schema`, and check the required fields in Swagger.

Reference

Security rules, answers to common questions, a knowledge check, a one-page cheat sheet, and definitions.

A Security do's and don'ts

✓ Do

- Prefer TID UUID + User API Secret for people, so changes are audited to them.
- Use client ID + secret only for unattended integrations, and limit who knows them.
- Practice write calls in B2WTechSupport.
- Hide tokens and secrets before sharing screenshots or logs.
- Clear secrets from Postman variables before sharing or exporting a collection.
- Coordinate before generating a new secret or API key that may be in use.

✗ Don't

- Never send a client ID and secret over email.
- Never put credentials in the URL or JSON body of a login call.
- Don't paste bearer tokens into tickets or chats. They work for a full day.
- Don't reset a password to fix a LoginWithTID 404. Fix the user link.
- Don't test changes in a customer environment without their request.
- Don't build new integrations on username and password. It is being retired.

B Frequently asked questions

Questions marked **From the training** were asked during the Ops API walkthrough.

What is the biggest difference between a client ID + secret and TID?

From the training



A client ID and secret give system-administrator-level access. TID keeps an audit trail of what was changed and takes on the persona of the user, following the limits set in their security role.

If someone has a read-only license but their security role lets them add records through the API, can they make changes? From the training ×

No. A read-only license does not have permission to make changes in Ops. Per the training, it cannot even run a GET on employees.

Will API security roles follow the organization, business unit, and user structure, so people only see their own data? From the training ×

Likely not. That would be a very large undertaking for the development team. Today, API permissions are set per area and per action (Create, View, Edit, Delete).

Can I change a record's ObjectID through the API? From the training ×

No. The ObjectID is the record's permanent key. You use it to target a PUT or DELETE, but you cannot change it.

What can you do with the API? From the training ×

Whatever you can do with the application, for the areas the API covers and within the caller's permissions. See the [endpoint explorer](#).

Is an AccessToken the same thing as a bearer token? ×

Yes. The login response calls it `AccessToken`; you send it in a header as `Authorization: Bearer <AccessToken>`.

How long does a token last? ×

One day by default, set by the `Security:JWT:Lifetime` site setting. Per the training, generating a new token also retires the old one, so use your latest.

Is username and password login going away? ×

Yes. It is going to sunset soon. Move people to TID UUID + User API Secret, and unattended integrations to client ID + secret.

Do I need to log in to use Ping or Version?



No. `/Ping/hello` and `/Version` are the two calls that need no authentication, which makes them the first check on any ticket. On B2WTechSupport, `/SystemInfo` also answers without a login.

Is the API URL case-sensitive?



No. `/OpsAPI_B2WTechSupport`, `/OPSAPI_B2WTechSupport`, and `/opsapi_b2wtechsupport` all reach the same API.

Where do I find the fields a POST needs?



Three places: the area's section in Swagger, `GET /{Area}/schema`, or a GET of an existing record in that area. The last one is often the quickest.

Why does Postman show an "Auth" folder if Auth isn't an endpoint?



It is only a folder that groups the four login requests. They all call `/Login` or `/LoginWithTID`.

Where can I read more?



The customer's own Swagger page, and the B2W Operational Suite API page on the cloud help site. Both are linked under [Links and resources](#).



Check your understanding

Eight scenarios based on real tickets. Pick an answer to see the explanation.

1. A customer's overnight import logs in fine, then `GET /Job` returns 401. What is the most likely cause?

- ☐ The environment name in the URL is wrong
- ☐ The bearer token is missing from the Authorization header or has expired
- ☐ The user's role lacks the Job permission
- ☐ The API site is down

2. Which address opens the Swagger catalog for B2WTechSupport?

- ☐ `.../B2WTechSupport/doc/index.html`
- ☐ `.../EstAPI_B2WTechSupport/doc/index.html`
- ☐ `.../OpsAPI_B2WTechSupport/doc/index.html`
- ☐ `.../OpsAPI/B2WTechSupport/swagger`

3. `LoginWithTID` returns 404. What should you do?

- ☐ Reset the user's password
- ☐ Fix the link between their Trimble ID and their Ops user record
- ☐ Generate a new client secret
- ☐ Wait 24 hours for the token to reset

4. A scheduled overnight integration has no person behind it. Which login method fits?

- ☐ Username and password
- ☐ A bearer token copied from Postman
- ☐ Client ID and client secret
- ☐ LoginWithTID

5. A GET returns exactly 100 records. What does that usually mean?

- ☐ The customer has exactly 100 records
- ☐ The call hit the page size, so request the next page with `$skip`
- ☐ The token is about to expire
- ☐ The `$filter` is wrong

6. The customer's request body starts with `<soap:Envelope>`. What are they using?

- ☐ The legacy SOAP Web Services
- ☐ Ops API 2.0
- ☐ The Estimate API
- ☐ Swagger

7. Where do credentials go when you call `GET /Login ?`

- ☐ In the URL query string
- ☐ In the JSON body
- ☐ Anywhere, as long as the names are right
- ☐ In HTTP headers

8. A user gets 403 “insufficient privileges” on `POST /Employee`. Where do you look first?

- ☐ The Cloud Management Portal
- ☐ Their security role: API section, Employee row, Create column
- ☐ Their bearer token
- ☐ The environment URL

D Quick reference

Everything on one screen. It prints on a single page if you print this section alone.

ADDRESSES

API	<code>https://{cluster}/OpsAPI_{environment}</code>
Docs	<code>{api}/doc/index.html</code>
Spec	<code>{api}/doc/v2/OpsAPI.json</code>
Example	<code>https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport</code>

NO LOGIN NEEDED

GET `/Ping/hello` → server name + your text

GET `/Version` → API version

GET `/SystemInfo` → TID mode (B2WTechSupport)

LOG IN · GET /LOGIN · HEADERS ONLY

TID `tiduuid` + `apiSecret` → acts as the user

Client `clientId` + `clientSecret` → System Administrator

User `userName` + `password` → being retired

Trimble `GET /LoginWithTID` with a Trimble ID token

DATA CALLS

Header `Authorization: Bearer <AccessToken>`

Body `Content-Type: application/json`

Token Lasts 1 day; a new login retires the old one

Query `$filter $select $orderby $top $skip` · 100 per page

WHERE CREDENTIALS LIVE

TID ID + secret ⚙ Users → User View → User API Secret

Client ID + secret ⚙ Integrations & Add-ins → Manage API Key

API URL Cloud Management Portal → API Configuration

FIVE QUESTIONS FOR EVERY TICKET

1. Ops API (JSON) or Web Services (SOAP XML)?
2. Right URL? Does `/Ping/hello` answer?
3. Which login method?
4. Login status and body, on their own?
5. Bearer token in the data call's header?

STATUS CODES

- 400** Bad input, or no credentials in the login headers
- 401** Credentials rejected, or token missing/expired
- 403** Security role lacks the API permission
- 404** Not found; on LoginWithTID, fix the user link

NEVER

- Email a client ID and secret
- Put credentials in a URL or body
- Paste tokens into tickets
- Reset a password for a LoginWithTID 404

E Glossary

AccessToken / bearer token

The signed token the login returns. Sent on every data call as `Authorization: Bearer ...`. Lasts 1 day by default.

API

Application programming interface: a published way for programs to request actions from another system.

apiSecret / User API Secret

A per-user secret generated in Ops. Paired with the TID ID to log in as that user.

Base URL

The root address of a customer's API, such as `https://b2w-eus10.b2w.trimble.com/OpsAPI_B2WTechSupport`.

Client ID / client secret

Environment-level credentials from Manage API Key. They log in as the built-in System Administrator.

Cluster

The server group that hosts a customer, such as `b2w-eus10.b2w.trimble.com`.

CRUD

Create, Read, Update, Delete: the four basic data actions, matching POST, GET, PUT, and DELETE.

Endpoint

One address and method the API answers, such as `GET /Employee`.

Environment

A customer's site name, such as `B2WTechSupport`.

Functional area

A group of related endpoints in the catalog, such as Employee or Job.

Header

A key-value pair sent with a request, such as `Content-Type: application/json`.

JSON

The plain-text data format the Ops API sends and receives.

JWT

JSON Web Token: the signed format of the AccessToken.

LoginWithTID

The endpoint that exchanges a Trimble ID token for an Ops access token.

ObjectID

A record's permanent unique key (a GUID). Used to target updates and deletes. It can't be changed.

OData

The query standard behind `$filter`, `$select`, `$orderby`, `$top`, and `$skip`.

Page size

The most records one GET returns: 100 by default.

Postman collection

A shared, ready-made set of requests and variables, imported into Postman.

REST

The API style the Ops API uses: one URL per resource, standard HTTP methods, JSON data.

RowVersion

A number on each record that changes when it is saved. Send it back unchanged on a PUT.

Security role

The Ops role whose API section grants Create, View, Edit, and Delete per area.

SOAP / Web Services

The older B2W API: `.asmx` URLs, POST-only, XML envelopes.

Status code

The three-digit result of a call, such as 200, 401, or 403.

Swagger

The interactive catalog at `/doc/index.html` that lists every endpoint and field.

TID (Trimble ID)

Trimble's identity service. In TID mode, users sign in to Ops with their Trimble ID.

tiduuid / TID ID

The unique ID of a user's Trimble identity, shown in the User API Secret dialog.

Links and resources



Download Postman

The free desktop app for Windows, macOS, and Linux.

[postman.com/downloads](https://www.postman.com/downloads)

(<https://www.postman.com/downloads/>)



B2W Ops API Postman collection

The team collection. Sign in with your Trimble Google account.

drive.google.com

(<https://drive.google.com/file/d/1ffhAF0z383mx1lo5qvFcjEHLsStrS9R/view?usp=sharing>)



B2WTechSupport API docs (Swagger)



OpenAPI file (OpsAPI.json)

Every endpoint, field, and code for our example environment.

(https://b2weus10.b2w.trimble.com/OpsAPI_B2WTechSupport/doc/index.html)

The machine-readable catalog, importable into Postman.

(https://b2weus10.b2w.trimble.com/OpsAPI_B2WTechSupport/doc/v2/OpsAPI.json)



B2WTechSupport ping

Quick connectivity check, no login needed.

[.../OpsAPI_B2WTechSupport/Ping/hello](https://b2weus10.b2w.trimble.com/OpsAPI_B2WTechSupport/Ping/hello)

(https://b2weus10.b2w.trimble.com/OpsAPI_B2WTechSupport/Ping/hello)



B2W Operational Suite API help page

On the B2W Trimble cloud help site. Requires a Trimble sign-in.

sites.google.com/trimble.com/b2wtrimblecloud

(<https://sites.google.com/trimble.com/b2wtrimblecloud/home/accounting-integrations-apis/b2w-operational-suite-api?authuser=0>)



B2W Ops API Help Guide

Built for the B2W Support Team. The Ops API guide comes from the Ops API Support Training and its walkthrough notes. Endpoint lists, sample responses, and live results come from the B2WTechSupport environment, checked October 2, 2026. Credentials and tokens in screenshots are hidden.